

# FROM MATURITY TO OLD AGE

When a star has finished burning hydrogen in its core, it will start burning its outer layers in a series of concentric shells. The star will expand as the source of heat moves outward and its outer layers cool. Stars with very low mass will eventually fade and cool; Sun-like stars will evolve into red giants; and high-mass stars will become supergiants. Once a star has used up all its available nuclear fuel, it will deflate, because there is no longer any power source to replace the energy lost from its surface. As it collapses, if it has enough mass, its helium core starts to burn and change into carbon. Once the fuel in its core is used up again, helium-shell burning begins in the star's atmosphere and the star expands. In very massive stars, this process is repeated until iron is produced. When a Sun-like star has used up all of its fuel, it will lose its outer atmosphere in a spectacular planetary nebula and collapse to become a white dwarf. A high-mass star will explode as a supernova and leave behind a neutron star or black hole.

star expands as hydrogen-shell burning occurs

star starts to collapse as hydrogen is used up

## LOW-MASS STAR

Once a star with a mass less than half that of the Sun has used up the hydrogen in its core, it will convert the hydrogen in its atmosphere to helium and collapse, just as in higher-mass stars. However, low-mass stars do not have enough mass for the temperature and pressure at its core to get high enough for helium burning to occur. These stars will just gradually fade as they cool.

## STAR NOW ON THE MAIN SEQUENCE

Stars spend the greatest proportion of their lives on the main sequence. The more massive the star, the shorter the period of time it will spend on the main sequence, since larger stars burn their fuel at a faster rate than smaller ones.

## SUN-LIKE STAR

When a Sun-like star exhausts the hydrogen in its core, hydrogen-shell burning begins and it becomes a red giant, often losing its outer layers to produce a planetary nebula. It eventually collapses, and the temperature and pressure at its core initiate helium-core burning. The star again expands as helium-shell burning occurs before finally collapsing to become a white dwarf that gradually fades to black.

star becomes a red giant as hydrogen-shell burning starts

## HIGH-MASS STAR

The higher the mass of the star, the more times it will expand and contract—its mass dictates the temperature of the core each time it contracts. Different elements are produced at each stage. If the star is massive enough, an iron core is formed, but elements heavier than iron cannot be formed within stellar cores. They are formed in supernova explosions that leave behind neutron stars or black holes.

supergiant star produces heavier elements through nuclear reactions



## MOSTLY MAIN-SEQUENCE STARS

About 90 percent of the visible stars in a typical view of the night sky are on the main sequence. This corresponds with the fact that most stars spend 90 percent of their life on the main sequence.